SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the importance of DOF in robot design and operation. Discuss how the number and type of DOFs affect a robot's movement.

Q.24 Define pneumatic and hydraulic actuators used in robotic systems. Explain their construction, working principles.

Q.25 Explain how analog and digital sensors are integrated into a robotic system. Describe the process of signal conditioning, analog-to-digital conversion (ADC)

Roll No. ....

4th Sem / Instrumentation & Control

Subject : Robotics and Automation

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which of the following is an input device for a robot

a) Motor b) Actuator

c) Sensor d) Wheel

Q.2 Robots are classified based on

a) Color and size

b) Shape and language

c) Configuration, control and application

d) Speed and sound

Q.3 A robot with 4 degrees of freedom can

a) Only move up and down

b) Perform 4 independent movements

c) Move in only 2 directions

d) Not rotate at all

Q.4 Which of the following is NOT a type of actuator

a) Electric motor b) Hydraulic cylinder

c) Pneumatic cylinder d) Microcontroller

Q.5 Pneumatic grippers are powered by

a) Water b) Compressed air

c) Electricity d) Solar energy

Q.6 Which of these sensors can detect light intensity

a) Ultrasonic sensor

b) IR sensor

c) LDR (Light Dependent Resistor)

d) Encoder

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Which type of sensor provides continuous output values.

Q.8 What type of robot typically operates in hospitals for surgeries or assistance

Q.9 In a 4 DOF robotic arm, how many independent motions can it perform.

Q.10 Signal uses in Hydraulic actuator.

Q.11 Servo motor drivers are typically used for

Q.12 Which type of gripper is suitable for handling ferrous metal parts

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 What is a robot, List the main components of a robot.

Q.14 Explain the two main types of degrees of freedom in robot joints.

Q.15 Explain the indirect kinematic of robot.

Q.16 What is the main function of an driver in a robot and also discuss the electric drive.

Q.17 Explain the active and passive types of gripper.

Q.18 Discuss the pneumatic types of actuator.

Q.19 Write the 4 application of robot in commercial.

Q.20 What is the difference between an analog sensor and a digital sensor.

Q.21 What is an DAC (Digital to Analog Converter) and why is it important in robotics.

Q.22 Explain the following terms, Resolution, accuracy precision, repeatability